

Mean-Field Theory and Landau-Ginzburg Approach

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Review of mean-field theories

Equivalent formulations of the *mean-field (MF) approximation*:

- 1 each degree of freedom interacts equally with all other degrees of freedom,
- 2 each degree of freedom interacts only with some average of all the others,

Main properties:

- easy to study (solution of self-consistent equations for some physical quantity);
- neglect spatial structure and fluctuations;
- comparing MF theories for apparently different systems, we observe similar critical behavior.

MF example I: Weiss molecular-field theory

Several ways of deriving the mean-field approximation for the (ferromagnetic) Ising model.

Magnetization M satisfies

$$M = \tanh(\beta(h + zJM)) \quad (1)$$

For $h \simeq 0$, expanding r.h.s. for $T \rightarrow T_c$,

$$M \approx \beta h + \beta zJM - \frac{\beta^3}{3} (h + zJM)^3 \quad (2)$$

and introducing $T_c = zJ$, $\tau = T/T_c$ and using at first order $\beta h \approx (1 - \tau^{-1}) M$,

$$h = (T - T_c) M + \frac{1}{3\beta} M^3 + O(\tau M^3). \quad (3)$$

MF example II: Van der Waals theory

The equation of state of an *ideal gas* is

$$pv = k_B T \quad (4)$$

with p, v pressure and volume per particle.

By adding non-ideal gas features:

- excluded volume b per particle is not available to the gas
 $\implies v \rightarrow v - b$;
- due to interaction between particles, effective pressure is larger
 $\implies p \rightarrow p + \frac{a}{v^2}$

Van der Waals *equation of state* of simple fluids is obtained:

$$\left(p + \frac{a}{v^2}\right)(v - b) = k_B T \quad \implies \quad p = \frac{k_B T}{v - b} - \frac{a}{v^2}. \quad (5)$$

MF example II: Van der Waals theory

Liquid-vapor transition: the function $p(v)$ has a flex point at T_c , where

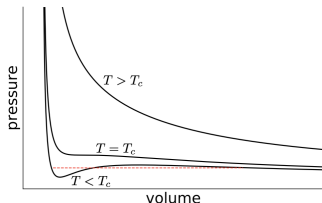
$$\left. \frac{\partial p}{\partial v} \right|_{T=T_c} = \left. \frac{\partial^2 p}{\partial v^2} \right|_{T=T_c} = 0 \quad (6)$$

it gives 2 conditions:

$$\frac{k_B T_c}{(v_c - b)^2} = \frac{2a}{v_c^3}, \quad \frac{2k_B T_c}{(v_c - b)^3} = \frac{6a}{v_c^4} \quad (7)$$
$$\implies v_c = 3b, \quad k_B T_c = \frac{8a}{27b}, \quad p_c = \frac{a}{27b^2}. \quad (8)$$

Introducing dim-less variables $\pi = p/p_c$, $\nu = v/v_c$ and $\tau = T/T_c$,

$$\pi = \frac{8\tau}{3\nu - 1} - \frac{3}{\nu^2} \quad (9)$$



MF example II: Van der Waals theory

Expanding for $T \rightarrow T_c$ in powers of $\eta = 1 - \nu$,

$$\pi = \frac{8\tau}{3(1-\eta) - 1} - \frac{3}{(1-\eta)^2} \quad (10)$$

$$= \frac{4\tau}{1 - \frac{3}{2}\eta} - \frac{3}{(1-\eta)^2} \quad (11)$$

$$\approx 4\tau \left[1 + \frac{3}{2}\eta + \frac{9}{4}\eta^2 + \frac{27}{8}\eta^3 \right] - 3 \left[1 + 2\eta + 3\eta^2 + 4\eta^3 \right] \quad (12)$$

$$= 1 + 4(\tau - 1) + 6(\tau - 1)\eta + 9(\tau - 1)\eta^2 + \frac{27}{2}\tau\eta^3 - 12\eta^3 \quad (13)$$

$$= 1 + 4t + 6t\eta + 9t\eta^2 + \frac{27}{2}t\eta^3 + \frac{3}{2}\eta^3 + \dots \quad (14)$$

where $t = (T - T_c)/T_c$.

MF example II: Van der Waals theory

For $t \rightarrow 0$, one can show that

$$\eta(t) \sim t^\beta, \quad \beta = \frac{1}{2} \quad (15)$$

which implies that $O(t\eta^2)$ terms can be neglected compared to $O(\eta^3)$ terms. It follows that

$$\tilde{\pi} = 6t\eta + \frac{3}{2}\eta^3 + O(t\eta^2, \eta^4) \quad (16)$$

where the quantity $\tilde{\pi} = \pi - 1 - 4t$ and η play the same role of h and M for the ferro-para transition.

Where does the analogy come from? Is there some *universal behavior* independent of the system's details?

Landau Theory: homogeneous systems

An **order parameter** m describes different physical phases by assuming value zero on one side of the critical point and non-zero value on the other.

Landau-Ginzburg **effective Hamiltonian** $\mathcal{H}_{LG}(m)$:

- ① consistent with the *symmetries* of the system;
- ② *analytic* function of m , can be expanded in series around criticality,

$$\mathcal{L}_{LG}(m) = \frac{\mathcal{H}_{LG}(m)}{V} = \sum_{k=0}^{\infty} \mathcal{L}_k m^k \quad (17)$$

- ③ in equilibrium, free-energy corresponds to the *minimum* of $\mathcal{H}_{LG}(m)$,

$$F = \min_m \mathcal{H}_{LG}(m) \quad (18)$$

and the order parameter m solves the mean-field equation, i.e.

$$M_0 = \begin{cases} 0 & T > T_c \\ f(T) & T < T_c \end{cases} \quad (19)$$

Landau Theory: homogeneous systems

Consider para-ferro phase transition, the Z_2 - symmetry requires $\mathcal{H}_{LG}(m) = \mathcal{H}_{LG}(-m)$, which gives the expansion

$$\mathcal{L}_{LG}(m) = \mathcal{L}_0 + \mathcal{L}_2 m^2 + \mathcal{L}_4 m^4 + \dots \quad (20)$$

Coefficients $\mathcal{L}_k = \mathcal{L}_k(T, \{J\})$:

- *non-universal* (depend on e.g. T , p , microscopic interactions)
- *analytic* functions of T , series expansion close to T_c :

$$\mathcal{L}_0(T, \{J\}) = \mathcal{L}_{0,0} + \mathcal{L}_{0,1}(T - T_c) + O((T - T_c)^2) \quad (21)$$

$$\mathcal{L}_2(T, \{J\}) = \mathcal{L}_{2,0} + \mathcal{L}_{2,1}(T - T_c) + O((T - T_c)^2) \quad (22)$$

$$\mathcal{L}_4(T, \{J\}) = \mathcal{L}_{4,0} + \mathcal{L}_{4,1}(T - T_c) + O((T - T_c)^2) \quad (23)$$

Landau Theory: homogeneous systems

- 1 high-temperature equilibrium solution $M_0 = 0$ implies $\beta\mathcal{H}_{LG}/V \approx \mathcal{L}_0(T, \{J\})$ for $T > T_c$, which is a regular contribution. We can set $\mathcal{L}_0 = 0$.
- 2 for a finite minimum of $\mathcal{H}_{LG}$ (otherwise $M_0 \rightarrow \pm\infty$),

$$\mathcal{L}_4(T, \{J\}) \xrightarrow{T \rightarrow T_c} \mathcal{L}_{4,0} > 0 \quad (24)$$

- 3 imposing equilibrium condition

$$\left. \frac{\partial \mathcal{H}_{LG}}{\partial m} \right|_{m=M_0} = 0 \quad \Longrightarrow \quad 2\mathcal{L}_2 M_0 + 4\mathcal{L}_4 M_0^3 = 0 \quad (25)$$

with solutions $M_0 = 0$ and $M_0 = \pm \left(-\frac{\mathcal{L}_2}{2\mathcal{L}_4}\right)^{1/2}$. Solutions must be continuous at T_c , hence $\mathcal{L}_{2,0} = 0$ and

$$\mathcal{L}_2 \approx \mathcal{L}_{2,1}(T - T_c) \quad (26)$$

Landau Theory: continuous phase transitions

Consider the effective Hamiltonian

$$\frac{\mathcal{H}_{LG}(m)}{V} = \frac{r_0 t}{2} m^2 + \frac{u_0}{4} m^4 \quad (27)$$

with $r_0 = 2T_c \mathcal{L}_{2,1}$ and $u_0 = 4\mathcal{L}_{4,0}$.

From the equilibrium equation $r_0 t m + u_0 m^3 = 0$

$$M_0 = \begin{cases} 0 & T > T_c \\ \pm \left(-\frac{r_0 t}{u_0}\right)^{1/2} & T \leq T_c \end{cases} \quad (28)$$

which implies for $T \rightarrow T_c^-$ a critical behavior of the order parameter

$$M_0 \propto (T - T_c)^\beta, \quad \beta = \beta_{MF} = \frac{1}{2} \quad (29)$$

Landau Theory: continuous phase transitions

From the free energy

$$F = \mathcal{H}_{LG}(M_0) = \begin{cases} 0 & T > T_c \\ -\frac{r_0^2 t^2}{4u_0} V & T \leq T_c \end{cases}$$

compute the *specific heat*

$$c_V = -\frac{T}{V} \frac{\partial^2 \mathcal{H}_{LG}(M_0)}{\partial T^2} = \begin{cases} 0 & T > T_c \\ \frac{r_0^2}{2u_0 T_c^2} T & T < T_c \end{cases}$$

which has a discontinuity, i.e. a critical behavior

$$c_V \propto (T - T_c)^{-\alpha}, \quad \alpha = \alpha_{MF} = 0 \quad (30)$$

Landau Theory: continuous phase transitions

Add a small external field $h \neq 0$,

$$\frac{\mathcal{H}_{LG}(m)}{V} = \frac{r_0 t}{2} m^2 + \frac{u_0}{4} m^4 - h m \quad (31)$$

The *isothermal susceptibility* is

$$\chi_T(h) = \left. \frac{\partial M_0}{\partial h} \right|_T \quad (32)$$

$$= \left. \frac{\partial h}{\partial M_0} \right|_T^{-1} = \frac{1}{r_0 t + 3u_0 M_0(h)^2} \Big|_{h=0} = \begin{cases} \frac{1}{r_0 t} & T > T_c \\ -\frac{1}{2r_0 t} & T < T_c \end{cases} \quad (33)$$

and the critical behavior for $T \rightarrow T_c^\pm$ is

$$\chi_T \propto (T - T_c^\pm)^{-\gamma^\pm}, \quad \gamma^\pm = \gamma_{MF} = 1 \quad (34)$$

At $t = 0$, the *critical isothermal* is given by

$$r_0 t M_0 + u_0 M_0^3 - h = 0 \implies h \propto M_0^\delta, \quad \delta = \delta_{MF} = 3 \quad (35)$$

Landau Theory: mean-field critical exponents

In summary, *mean-field critical exponents* from Landau's theory are

$$\beta_{MF} = \frac{1}{2}, \quad \alpha_{MF} = 0, \quad \gamma_{MF} = 1, \quad \delta_{MF} = 3$$

REMARK: rationality of the exponents is a consequence of series expansion (true exponents are not)

REMARK: correlation length exponent ν not defined for homogeneous systems

REMARK: agreement with Weiss theory of ferromagnetism and Van der Waals theory of simple fluids.

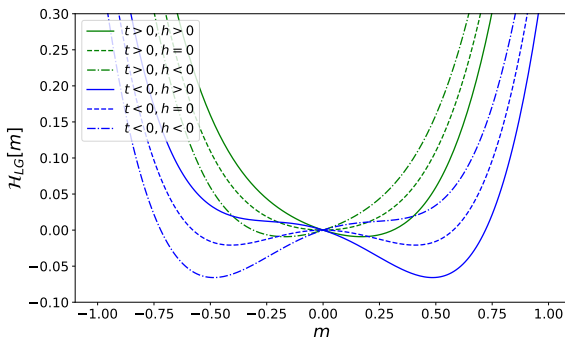
First signature of ***universality*** (different systems belong to the same *universality class*).

Landau Theory: discontinuous phase transitions

Consider the effective Hamiltonian with finite external field

$$\frac{\mathcal{H}_{LG}}{V} = \frac{r_0 t}{2} m^2 + \frac{u_0}{4} m^4 - h m \quad (36)$$

Stationary condition $r_0 t M_0 + u_0 M_0^3 - h = 0$ implies an $M_0[h] \neq 0$.



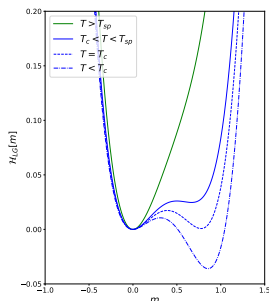
Landau Theory: discontinuous phase transitions

Suppose $h = 0$ but there is a third-order term in the effective Hamiltonian

$$\frac{\mathcal{H}_{LG}}{V} = \frac{r_0 t}{2} m^2 + \frac{u_0}{4} m^4 + \frac{C_3}{3} m^3 \implies r_0 t M_0 + u_0 M_0^3 + C_3 M_0^2 = 0 \quad (37)$$

Solutions: $M_0 = 0$, $M_{0,\pm} = -\frac{C_3}{2u_0} \pm \sqrt{\left(\frac{C_3}{2u_0}\right)^2 - \frac{r_0 t}{u_0}}$:

- $M_0 = 0$ always (local/global minimum)
- $M_{0,\pm}$ for $t < t_{sp} = T_{sp} - T_c = \frac{C_3^2}{4r_0 u_0}$ (e.g. if $C_3 < 0$, $M_{0,+}$ local minimum for $T_c < T < T_{sp}$)
- $M_{0,+}$ abruptly becomes the equilibrium at $T = T_c$ where $\mathcal{H}_{LG}(M_{0,+}) = \mathcal{H}_{LG}(0)$.



REMARK: Landau theory can be used to expand around metastable states.

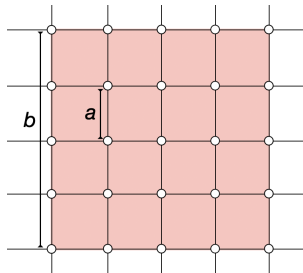
Landau Theory: inhomogeneous systems

For inhomogeneous systems, we have an *order parameter field* $m(\vec{x})$, defined on every point $\vec{x}$ of (a region of) $\mathbb{R}^d$.

It represents a local average over microscopic degrees of freedom.

Suppose we perform coarse-graining over region of scale b :

- large enough to have smooth field, i.e. $b \gg a$ (lattice spacing);
- small enough for fluctuations to play a role, i.e. $b \ll \xi(T)$;
- close to criticality, $\xi(T) \rightarrow \infty$, so any finite b should be ok.



REMARK: No well-defined coarse-graining procedure, theory weakly depends on b .

Landau Theory: inhomogeneous systems

In order to account for the *energetic cost of spatial inhomogeneities*, include spatial derivatives in the effective Hamiltonian

$$\nabla m(\vec{x}), m\nabla m(\vec{x}), \nabla^2 m(\vec{x}), (\nabla m(\vec{x}))^2, m\nabla^2 m, \dots \quad (38)$$

REMARK:

- the effective Hamiltonian is now a *functional*;
- terms must be symmetric w.r.t. $m \rightarrow -m$;
- only low-order terms are considered (long wavelength contributions);
- some terms are equivalent, e.g.

$$\int_V (m\nabla^2 m) d^d x = \int_{\partial V} (m\nabla m) \cdot \vec{u}_n d\Sigma - \int_V (\nabla m)^2 d^d x \quad (39)$$

Landau-Ginzburg effective Hamiltonian takes the minimal form

$$\mathcal{H}_{LG}[m(\vec{x})] \propto \int d\vec{x} \left(\frac{c_0}{2} (\nabla m(\vec{x}))^2 + \frac{r_0 t}{2} m(\vec{x})^2 + \frac{u_0}{4} m(\vec{x})^4 \right) \quad (40)$$

Microscopic derivation: coarse-graining

Is it possible to derive the expression of the effective Hamiltonian from a microscopic model?

Consider the Ising model on a d -dimensional hypercubic lattice of L^d sites with spacing a ,

$$H = -\frac{1}{2} \sum_{i \neq j} J_{ij} s_i s_j. \quad (41)$$

Coarse-graining over (non-overlapping) blocks B_i of linear size b around site i ,

$$m_{B_i}(\vec{x}_i) = \left(\frac{a}{b}\right)^d \sum_{j \in B_i} s_j = \frac{1}{n_B} \sum_{j \in B_i} s_j \quad (42)$$

REMARK: The coarse-grained magnetization $m_{B_i} \in [-1, 1]$ and dimensionless.

Microscopic derivation: partition function

$$Z = \sum_{\vec{s}} e^{-\beta H(\vec{s})} \quad (43)$$

$$= \sum_{\{m_B\}_{B=1}^{N_B}} \sum_{\vec{s}} e^{-\beta H(\vec{s})} \prod_{B=1}^{N_B} \delta_{n_B m_B, \sum_{j \in B} s_j} \quad (44)$$

$$\approx \sum_{\{m_B\}_{B=1}^{N_B}} e^{-\beta \tilde{H}[\{m_B\}]} \prod_{B=1}^{N_B} \sum_{\{s_j\}_{j \in B}} \delta_{n_B m_B, \sum_{j \in B} s_j} \quad (45)$$

$$= \sum_{\{m_B\}_{B=1}^{N_B}} e^{-\beta \tilde{H}[\{m_B\}] + S[\{m_B\}]} = \sum_{\{m_B\}_{B=1}^{N_B}} e^{-\beta \mathcal{H}_{\text{eff}}[\{m_B\}]}, \quad (46)$$

REMARK: non-overlapping blocks make energetic term complicated (higher-order interactions), overlapping blocks make entropic term complicated (double-counting problems).

Microscopic derivation: energetic term

$$\sum_{ij} J_{ij} s_i s_j \approx \sum_B m_B^2 \sum_{i,j \in B} J_{ij} + \sum_B \sum_{B' \neq B} m_B m_{B'} \sum_{i \in B, j \in B'} J_{ij} \quad (47)$$

$$\approx n_B^2 \sum_B m_B^2 J_B + n_B^2 \sum_B \sum_{B' \neq B} m_B m_{B'} \tilde{J}_B \quad (48)$$

Approximating $\sum_B \rightarrow b^{-d} \int d\vec{x}$, the continuum limit,

$$\rightarrow \frac{b^d J_B}{a^{2d}} \int d\vec{x} m(\vec{x})^2 + \frac{1}{a^{2d}} \int d\vec{x} d\vec{x}' \tilde{J}(\vec{x} - \vec{x}') m(\vec{x}) m(\vec{x}') \quad (49)$$

$$\approx \frac{1}{a^{2d}} \int d\vec{x} d\vec{x}' \tilde{J}(|\vec{x} - \vec{x}'|) m(\vec{x}) m(\vec{x}') \quad (50)$$

where $\tilde{J}(|\vec{x} - \vec{x}'|)$ is an effective interaction between blocks (and we neglected intra-block terms). For b not too large, $\tilde{J} \simeq J$.

Microscopic derivation: energetic term

Reordering terms

$$\tilde{H} = - \int \frac{d\vec{x}d\vec{x}'}{2a^{2d}} J(|\vec{x} - \vec{x}'|) m(\vec{x})m(\vec{x}') \quad (51)$$

$$= \int \frac{d\vec{x}d\vec{x}'}{4a^{2d}} J(|\vec{x} - \vec{x}'|) \left[(m(\vec{x}) - m(\vec{x}'))^2 - m(\vec{x})^2 - m(\vec{x}')^2 \right] \quad (52)$$

$$= \int \frac{d\vec{x}d\vec{x}'}{4a^{2d}} J(|\vec{x} - \vec{x}'|) (m(\vec{x}) - m(\vec{x}'))^2 \\ - \frac{1}{2a^{2d}} \int d\vec{r} J(r) \int d\vec{x} m(\vec{x})^2 \quad (53)$$

and for $J(|\vec{x} - \vec{x}'|)$ short-ranged, $m(\vec{x}') - m(\vec{x}) \approx \nabla m(\vec{x}) \cdot (\vec{x}' - \vec{x})$,

$$\tilde{H} \simeq \frac{1}{4a^{2d}} \int d\vec{r} d\vec{x} J(r) (\nabla_{\vec{x}} m(\vec{x}) \cdot \vec{r})^2 \\ - \frac{1}{2a^{2d}} \int d\vec{r} J(r) \int d\vec{x} m(\vec{x})^2. \quad (54)$$

Microscopic derivation: energetic term

Because of symmetries, all cross-products of type $\frac{\partial m}{\partial x_i} \frac{\partial m}{\partial x_j} r_i r_j$ vanish after integration, so we have just

$$\begin{aligned} \tilde{H} \simeq & \frac{1}{4a^{2d}} \sum_{i=1}^d \left(\int d\vec{r} J(r) r_i^2 \right) \left(\int d\vec{x} \left(\frac{\partial m}{\partial x_i} \right)^2 \right) \\ & - \frac{1}{2a^{2d}} \left(\int d\vec{r} J(r) \right) \left(\int d\vec{x} m(\vec{x})^2 \right). \end{aligned} \quad (55)$$

Defining

$$J_0 = \int d^d r J(r), \quad (56)$$

$$J_2 = \frac{1}{2} \int d^d r J(r) r_i^2 = \frac{1}{2d} \int d^d r J(r) r^2, \quad (57)$$

the latter valid for isotropic interactions. Finally,

$$\tilde{H}[m] = \frac{J_2}{2a^{2d}} \int d^d x (\nabla m)^2 - \frac{J_0}{2a^{2d}} \int d^d x m(\vec{x})^2. \quad (58)$$

Microscopic derivation: entropic term

The number of ways of generating magnetization m_{B_i} inside block B_i is

$$\mathcal{N}_{B_i} = \frac{n_B!}{n_i^+! n_i^-!} \quad (59)$$

with $n_i^\pm = n_B \frac{1 \pm m_{B_i}(\vec{x}_i)}{2}$. Using Stirling's approximation,

$$\mathcal{N}_{B_i} \approx \frac{n_B^{n_B} e^{-n_B}}{\prod_{s=\pm 1} \left(n_B \frac{1+sm_{B_i}}{2} \right)^{n_B \frac{1+sm_{B_i}(\vec{x}_i)}{2}} e^{-n_B \frac{1+sm_{B_i}(\vec{x}_i)}{2}}} \quad (60)$$

$$= \frac{n_B^{n_B} e^{-n_B}}{n_B^{n_B} e^{-n_B} \prod_{s=\pm 1} \left(\frac{1+sm_{B_i}}{2} \right)^{n_B \frac{1+sm_{B_i}(\vec{x}_i)}{2}} e^{-n_B sm_{B_i}(\vec{x}_i)/2}} \quad (61)$$

$$= \left[\left(\frac{1+m_{B_i}(\vec{x}_i)}{2} \right)^{\frac{1+m_{B_i}(\vec{x}_i)}{2}} \left(\frac{1-m_{B_i}(\vec{x}_i)}{2} \right)^{\frac{1-m_{B_i}(\vec{x}_i)}{2}} \right]^{-n_B} \quad (62)$$

Microscopic derivation: entropic term

The block entropy $S_{B_i} = \log \mathcal{N}_{B_i}$ is

$$S_{B_i}(\vec{x}_i) \approx n_B \left\{ \log 2 - \frac{1 + m(\vec{x})}{2} \log(1 + m(\vec{x})) - \frac{1 - m(\vec{x})}{2} \log(1 - m(\vec{x})) \right\} \quad (63)$$

$$\approx \left(\frac{b}{a}\right)^d \left\{ \log 2 - \frac{1}{2} m(\vec{x})^2 - \frac{1}{12} m(\vec{x})^4 \right\} \quad (64)$$

Because of extensivity, $S = \sum_{B_i} S_{B_i}$, and taking the continuous limit,

$$S = \frac{1}{b^d} \int d\vec{x} S(\vec{x}) \quad (65)$$

$$\approx -\frac{1}{a^d} \int d\vec{x} \left\{ \frac{1}{2} m(\vec{x})^2 + \frac{1}{12} m(\vec{x})^4 \right\} + \text{const.} \quad (66)$$

Microscopic derivation: functional $\mathcal{H}_{\text{eff}}$

Up to irrelevant constants, the effective Hamiltonian is given by

$$\mathcal{H}_{\text{eff}}[m] = \tilde{H}[\{m_B\}] - \frac{1}{\beta} S[\{m_B\}] \quad (67)$$

$$\begin{aligned} \approx & \frac{J_2}{2a^{2d}} \int d^d x (\nabla m)^2 + \left[\frac{1}{2\beta a^d} - \frac{J_0}{2a^{2d}} \right] \int d^d x m(\vec{x})^2 \\ & + \frac{1}{12\beta a^d} \int d^d x m(\vec{x})^4 \end{aligned} \quad (68)$$

$$\begin{aligned} = & \frac{J_2}{2a^{2d}} \int d^d x (\nabla m)^2 + \frac{J_0}{2a^{2d}} \left[\frac{k_B T a^d}{J_0} - 1 \right] \int d^d x m(\vec{x})^2 \\ & + \frac{1}{12\beta a^d} \int d^d x m(\vec{x})^4 \end{aligned} \quad (69)$$

REMARK: The critical temperature is $T_0 = J_0 a^{-d} / k_B$.

REMARK: No explicit dependence on b , it is hidden in the parameters.

Microscopic derivation: functional $\mathcal{H}_{\text{eff}}$

Defining the parameters $\tau = T/T_0$

$$c_0 = \frac{\beta J_2}{a^{2d}}, \quad r_0 = \frac{\beta J_0}{a^{2d}}, \quad u_0 = \frac{1}{3a^d} = \frac{\beta \tau J_0}{3a^{2d}}, \quad t = \tau - 1 \quad (70)$$

we finally obtain a non-dimensional functional of the local field

$$\beta \mathcal{H}_{\text{eff}}[m] = \int d^d x \left[\frac{c_0}{2} (\nabla m)^2 + \frac{r_0 t}{2} m(\vec{x})^2 + \frac{u_0}{4} m(\vec{x})^4 \right] \quad (71)$$

and the partition function becomes a *functional integral*

$$Z = \int Dm e^{-\beta \mathcal{H}_{\text{eff}}[m]} \quad (72)$$

Functionals and functional integrals

A *functional* is a mapping from a space of functions into real (or complex) numbers.

DEFINITION: Consider a function $f : x_0 \mapsto f(x_0)$, a functional $F[f]$ is a mapping $f \mapsto F[f]$ of the function f to its value at a point x_0 (or set of points).

REMARK: Here f is the argument of the functional, x_0 is a parameter.

EXAMPLE: The work done on a physical point by the force $\vec{f}$,

$$W[f] = \int_{x_0}^{x_1} \vec{f}(\vec{x}) \cdot d\vec{s}$$

is a functional of the force $\vec{f}$ (integrated over a displacement).

EXAMPLE: The effective Hamiltonian $\mathcal{H}_{\text{eff}} : m \mapsto \mathcal{H}_{\text{eff}}[m]$ is a functional of the order parameter $m : \vec{x} \mapsto m(\vec{x})$.

Functionals and functional integrals

DEFINITION: A *functional integral* is the integral of a functional $F[f]$ over its argument f .

Suppose that f is evaluated over a finite set of N points, in the domain of f , consider the N -dimensional Riemann integral over a set of parameters $\{f_i\}$. The functional integral is obtained as the limit $N \rightarrow \infty$,

$$\int Df F[f] \equiv \int \prod_{\vec{x}} df(\vec{x}) F[f] \quad (73)$$

$$= \lim_{N \rightarrow \infty} \int \prod_{i=1}^N df(\vec{x}_i) F[f] \quad (74)$$

$$= \lim_{N \rightarrow \infty} \int \prod_{i=1}^N df_i F[\{f_i\}] \quad (75)$$

Functionals and functional integrals

EXAMPLE: Consider the functional $F[f] = \int_0^L dx (f(x))^2$ and the functional integral

$$\mathcal{I} = \int Df e^{-\beta F[f]} = \int Df e^{-\beta \int_0^L dx f^2(x)}. \quad (76)$$

Discretizing the domain into a finite set $S = \left\{ \frac{1}{N}, \frac{2}{N}, \dots, 1 \right\}$ with $x_i = \frac{L}{N}i$ and $f_i = f(x_i)$, we obtain

$$I_N = \int df_1 df_2 \dots df_N e^{-\frac{L\beta}{N} \sum_{i=1}^N f_i^2} \quad (77)$$

$$= \prod_{i=1}^N \int df_i e^{-\frac{L\beta}{N} f_i^2} \quad (78)$$

$$= \left(\frac{\pi N}{L\beta} \right)^{\frac{N}{2}} \quad (79)$$

which is computed as N independent Gaussian integrals.

Functionals and functional integrals

REMARK: The integral I_N diverges as $N \rightarrow \infty$, unless redefine Df so that

$$df_i \rightarrow d\tilde{f}_i = df_i \sqrt{\frac{\beta L}{\pi N}}. \quad (80)$$

However, in the Riemann integral, the limit

$$S_N = \sum_{i=0}^{N-1} f(\hat{x}_i) (x_{i+1} - x_i) \rightarrow \int dx f(x) \quad (81)$$

exists because the $(x_{i+1} - x_i) \rightarrow dx$ is independent of f . Here, the discretization step $a_0 = L/N$ is independent of f but to keep finite the integral we should divide by

$$\mathcal{N}(a_0, \beta) = \left(\frac{\beta a_0}{\pi} \right)^{\frac{N}{2}} \quad (82)$$

which may depend on the argument of the functional.

Partition function as a functional integral

The original sum over microscopic states first becomes a sum over local magnetization values in a finite volume partition, then a functional integral taking the continuous limit, i.e.

$$Z = \lim_{N_B \rightarrow \infty} \lim_{a \rightarrow 0} \sum_{\{m_{B_i}(\vec{x}_i)\}_{i=1}^{N_B}} e^{-\beta \mathcal{H}_{\text{eff}}[\{m_{B_i}(\vec{x}_i)\}_{i=1}^{N_B}]} = \int Dm e^{-\beta \mathcal{H}_{\text{eff}}[m]}, \quad (83)$$

with the functional measure

$$Dm = \lim_{N_B \rightarrow \infty} \lim_{a \rightarrow 0} \int \prod_{i=1}^{N_B} dm(\vec{x}_i) \quad (84)$$

REMARK: The measure normalization can diverge in the limit, however real systems are discrete and all relevant physical quantities are ratios of functional integrals.

Partition function as a functional integral

If the functional integral can be expressed in terms of a suitable parametrized family of functions, i.e. $f(x) = \sum_n c_n \phi_n(x)$, the integration is then performed in the corresponding parameters space,

$$\int Df F[f] \equiv \int \prod_i dc_i F[\{c_n\}_{n=1}^{\infty}; \{\phi_n\}_{n=1}^{\infty}] \quad (85)$$

In this way, it is simpler to control the appearance of divergences.

For translational invariant systems, we can exploit the Fourier decomposition of the order parameter.

Partition function as a functional integral

For a finite number of degrees of freedom, we have a discrete, finite set of Fourier components, with upper *momentum cutoff* $\Lambda = \pi/b$ (or $\Lambda \propto 1/a$).

If the field $m(\vec{x})$ is real, then

$$\hat{m}(-\vec{k}) = \hat{m}^*(\vec{k}) \implies \begin{cases} \Re \hat{m}(\vec{k}) = \Re \hat{m}(-\vec{k}) \\ \Im \hat{m}(\vec{k}) = -\Im \hat{m}(-\vec{k}) \end{cases} \quad (86)$$

Integration should be performed on half of the momentum space to avoid double counting. Say we choose one axis k_x , we have

$$\int Dm = \lim_{\Lambda \rightarrow \infty} \int \prod_{|\vec{k}| < \Lambda, k_x > 0} d\hat{m}(\vec{k}) d\hat{m}^*(\vec{k}), \quad (87)$$

Review of Fourier Transform

Given a finite volume $V = L^d$, we define the Fourier transform $\hat{m}(\vec{k})$ of a (Lebesgue-measurable) function $m(\vec{x})$ by

$$\hat{m}(\vec{k}) = \mathcal{F}[m] = \int_V d\vec{x} m(\vec{x}) e^{-i\vec{k} \cdot \vec{x}}, \quad (88)$$

where the wavevector or momentum $\vec{k} = \frac{2\pi}{L} \vec{n}$ with $\vec{n} \in \mathbb{Z}^d$.

Its inverse operation is

$$m(\vec{x}) = \mathcal{F}^{-1}[\hat{m}] = \frac{1}{V} \sum_{\vec{k}} \hat{m}(\vec{k}) e^{i\vec{k} \cdot \vec{x}}. \quad (89)$$

In the infinite volume limit, $\sum_{\vec{k}} \rightarrow \frac{V}{(2\pi)^d} \int d^d k$, i.e.

$$m(\vec{x}) = \int \frac{d\vec{k}}{(2\pi)^d} \hat{m}(\vec{k}) e^{i\vec{k} \cdot \vec{x}} \quad (90)$$

Review of Fourier Transform

REMARK: plugging the inverse-transform expression of $\hat{m}(\vec{k})$ into that of the Fourier transform,

$$\begin{aligned} m(\vec{x}) &= \frac{1}{V} \sum_{\vec{k}} \hat{m}(\vec{k}) e^{i\vec{k} \cdot \vec{x}} \\ &= \frac{1}{V} \sum_{\vec{k}} \int_V d\vec{x}' m(\vec{x}') e^{i\vec{k} \cdot (\vec{x} - \vec{x}')} \\ &= \int_V d\vec{x}' m(\vec{x}') \left[V^{-1} \sum_{\vec{k}} e^{i\vec{k} \cdot (\vec{x} - \vec{x}')} \right] \end{aligned} \quad (91)$$

From the identity $m(\vec{x}) = \int_V d\vec{x}' m(\vec{x}') \delta(\vec{x} - \vec{x}')$, we find the Fourier transform of the Dirac delta function

$$\delta(\vec{x} - \vec{x}') = \frac{1}{V} \sum_{\vec{k}} e^{i\vec{k} \cdot (\vec{x} - \vec{x}')} \xrightarrow{V \rightarrow \infty} \int \frac{d\vec{k}}{(2\pi)^d} e^{i\vec{k} \cdot (\vec{x} - \vec{x}')} \quad (92)$$

Review of Fourier Transform

REMARK: plugging the Fourier transform expression of $m(\vec{x})$ into that of the inverse transform,

$$\begin{aligned}\hat{m}(\vec{k}) &= \int_V d\vec{x} m(\vec{x}) e^{-i\vec{k}\cdot\vec{x}} \\ &= \int_V d\vec{x} \frac{1}{V} \sum_{\vec{k}'} \hat{m}(\vec{k}') e^{-i(\vec{k}-\vec{k}')\cdot\vec{x}} \\ &= \sum_{\vec{k}'} \hat{m}(\vec{k}') \left[\frac{1}{V} \int_V d\vec{x} e^{-i(\vec{k}-\vec{k}')\cdot\vec{x}} \right]\end{aligned}\quad (93)$$

From the identity $\hat{m}(\vec{k}) = \sum_{\vec{k}'} \hat{m}(\vec{k}') \delta_{\vec{k},\vec{k}'}$, we find the Fourier transform of the Kronecker delta

$$\delta_{\vec{k},\vec{k}'} = \frac{1}{V} \int_V d\vec{x} e^{-i(\vec{k}-\vec{k}')\cdot\vec{x}} \xrightarrow[V \rightarrow \infty]{} \delta(\vec{k} - \vec{k}') = \int \frac{d\vec{x}}{(2\pi)^d} e^{-i(\vec{k}-\vec{k}')\cdot\vec{x}}. \quad (94)$$

Dimensional analysis of the effective Hamiltonian

Suppose we consider Landau's phenomenological approach,

$$\beta \mathcal{H}_{\text{eff}}[m] = \int d^d x \left[\frac{c_0}{2} (\nabla m)^2 + \frac{r_0 t}{2} m(\vec{x})^2 + \frac{u_0}{4} m(\vec{x})^4 \right] \quad (95)$$

Dimensionally, $[\beta \mathcal{H}_{\text{eff}}] = 1$ implies

$$[c_0] [m]^2 = [L]^{2-d} = \Lambda^{d-2}, \quad [r_0 t] [m]^2 = [u_0] [m]^4 = [L]^{-d} = \Lambda^d, \quad (96)$$

which admits several possible solutions. If we impose $[m] = 1$, then

$$[c_0] = \Lambda^{d-2}, \quad [r_0 t] = \Lambda^d, \quad [u_0] = \Lambda^d \quad (97)$$

If we rescale the field $m = a^{d/2} \phi$, we get

$$\beta \mathcal{H}_{\text{eff}}[\phi] = \int d^d x \left[\frac{c}{2} (\nabla \phi)^2 + \frac{rt}{2} \phi(\vec{x})^2 + \frac{u}{4} \phi(\vec{x})^4 \right] \quad (98)$$

$$[\phi] = \Lambda^{d/2}, \quad [c] = \Lambda^{-2}, \quad [rt] = 1, \quad [u] = \Lambda^{-d}. \quad (99)$$

Dimensional analysis of the effective Hamiltonian

It is also common to choose the *canonical scaling* by defining $\varphi(\vec{x}) = c^{1/2}\phi(\vec{x})$, and introducing new parameters

$$\mu^2 = \frac{rt}{c}, \quad \lambda = 3! \frac{u}{c^2}, \quad (100)$$

which gives

$$\beta \mathcal{H}_{\text{eff}}[\varphi] = \int d^d x \left[\frac{1}{2} (\nabla \varphi)^2 + \frac{\mu^2}{2} \varphi(\vec{x})^2 + \frac{\lambda}{4!} \varphi(\vec{x})^4 \right] \quad (101)$$

where

$$[\varphi] = \Lambda^{d/2-1}, \quad [\mu^2] = \Lambda^2, \quad [\lambda] = \Lambda^{4-d}. \quad (102)$$

REMARK: Notice that $\mu^2 \sim J_0/J_2$, i.e. it represents the inverse spatial range of the microscopic interaction (large “mass” means small extension).

General field-theoretic formalism: the Action

In the framework provided by (statistical and quantum) field theory, a quantity like $\beta\mathcal{H}_{\text{eff}}$ is called the **action** $S[\varphi]$, with general form

$$S[\varphi] = \int d\vec{x} \left[\frac{1}{2} (\nabla\varphi)^2 + g_1\varphi(\vec{x}) + \frac{g_2}{2}\varphi(\vec{x})^2 + \cdots + \frac{g_n}{n!}\varphi(\vec{x})^n + \cdots \right],$$

where

$$[\varphi] = \Lambda^{d/2-1}, \quad [g_1] = [\Lambda^{d/2+1}], \quad [g_2] = [\mu^2] = \Lambda^2, \quad [g_4] = [\lambda] = \Lambda^{4-d} \quad (103)$$

and the coefficient of the n -th order term has dimension

$$[g_n] = \Lambda^{-\delta_n}, \quad \delta_n = \frac{nd}{2} - d - n. \quad (104)$$

General field-theoretic formalism: the Action

The condition $\delta_n \geq 0$ identifies terms whose coupling constant is negligibly weak when $\Lambda \rightarrow +\infty$. It gives

$$d \geq d_n = \frac{2n}{n-2} \quad (105)$$

which is the dimension above which n -th order terms become irrelevant in the continuum limit.

REMARK: The dimension d_{uc} at which the lowest $n > 2$ term becomes irrelevant is the *upper critical dimension*.

REMARK: Dimensional counting gives information only on the ultraviolet (UV) divergences, not on the infrared (IR) ones, associated with the large-scale behavior of the system (emerging in the critical region).

Functional derivatives

If $F[m]$ is a functional of the field $m(\vec{x})$, the *functional derivative* of F with respect to $m(\vec{x})$ is defined as

$$\frac{\delta F[m]}{\delta m(\vec{x})} = \lim_{\epsilon \rightarrow 0} \frac{F[m(\vec{y}) + \epsilon \delta(\vec{y} - \vec{x})] - F[m(\vec{y})]}{\epsilon}. \quad (106)$$

EXAMPLE: Show that

$$\frac{\delta}{\delta m(\vec{x})} \int d\vec{y} m(\vec{y}) = \int \delta(\vec{y} - \vec{x}) d\vec{y} \quad (107)$$

Applying the definition

$$\begin{aligned} \frac{\delta}{\delta m(\vec{x})} \int d\vec{y} m(\vec{y}) &= \lim_{\epsilon \rightarrow 0} \frac{\int d\vec{y} (m(\vec{y}) + \epsilon \delta(\vec{y} - \vec{x})) - \int d\vec{y} m(\vec{y})}{\epsilon} \\ &= \int \delta(\vec{y} - \vec{x}) d\vec{y} \end{aligned} \quad (108)$$

Functional derivatives

Properties of functional derivatives:

- linear combination of functionals F and G

$$\frac{\delta}{\delta m(\vec{x})} (\alpha F[m] + \beta G[m]) = \alpha \frac{\delta F[m]}{\delta m(\vec{x})} + \beta \frac{\delta G[m]}{\delta m(\vec{x})}; \quad (109)$$

- product of functionals F and G

$$\frac{\delta}{\delta m(\vec{x})} (F[m] G[m]) = G[m] \frac{\delta F[m]}{\delta m(\vec{x})} + F[m] \frac{\delta G[m]}{\delta m(\vec{x})}; \quad (110)$$

- composition $F[m] = f(G[m])$

$$\frac{\delta}{\delta m(\vec{x})} (F[m]) = f'(G[m]) \frac{\delta G[m]}{\delta m(\vec{x})}; \quad (111)$$

- chain rule with $F[h] = F[h[m]]$, in which $h(\vec{x})$ is implicitly a functional of $m(\vec{x})$,

$$\frac{\delta F}{\delta m(\vec{x})} = \int d\vec{y} \frac{\delta F[h(\vec{y})]}{\delta h(\vec{y})} \frac{\delta h(\vec{y})}{\delta m(\vec{x})}. \quad (112)$$

EXERCISE: Show that

$$\frac{\delta}{\delta m(\vec{x})} \int d\vec{y} f(\vec{y}) m(\vec{y})^p = p f(\vec{x}) m(\vec{x})^{p-1},$$

$$\frac{\delta}{\delta m(\vec{x})} \int d\vec{y} V[m(\vec{y})] = V'[m(\vec{x})],$$

$$\frac{\delta}{\delta m(\vec{x})} \int d\vec{y} (\nabla m(\vec{y}))^2 = -2\nabla^2 m(\vec{x}).$$

Functional derivatives

REMARK: The functional $\mathcal{H}_{\text{eff}}[m] = \int d^d x \mathcal{L}(m(\vec{x}), \nabla m(\vec{x}))$ satisfies the *Euler-Lagrange equation*

$$\frac{\delta \mathcal{H}_{\text{eff}}}{\delta m(\vec{x})} = \frac{\partial \mathcal{L}}{\partial m} - \nabla \cdot \frac{\partial \mathcal{L}}{\partial \nabla m} \quad (113)$$

which can be proved using again the definition

$$\begin{aligned} \frac{\delta \mathcal{H}_{\text{eff}}}{\delta m(\vec{x})} &= \lim_{\epsilon \rightarrow 0} \left[\epsilon^{-1} \int d^d y (\mathcal{L}(m(\vec{y}) + \epsilon \delta(\vec{y} - \vec{x}), \nabla m(\vec{y}) + \epsilon \nabla \delta(\vec{y} - \vec{x})) \right. \\ &\quad \left. - \mathcal{L}(m(\vec{y}), \nabla m(\vec{y})) \right] \\ &= \int d^d y \frac{\partial \mathcal{L}(m, \nabla m)}{\partial m(\vec{y})} \delta(\vec{y} - \vec{x}) + \int d^d y \frac{\partial \mathcal{L}(m, \nabla m)}{\partial \nabla m(\vec{y})} \cdot \nabla \delta(\vec{y} - \vec{x}) \\ &= \frac{\partial \mathcal{L}(m, \nabla m)}{\partial m(\vec{x})} - \nabla \cdot \frac{\partial \mathcal{L}(m, \nabla m)}{\partial \nabla m(\vec{x})}. \end{aligned} \quad (114)$$

Physical observables via functional derivatives

Given the partition function

$$Z[h] = \int Dm e^{-\beta \mathcal{H}_{\text{eff}}[m] + \beta \int d^d x h(\vec{x}) m(\vec{x})} \quad (115)$$

$$= e^{-\beta F[h]} \quad (116)$$

where $F[h]$ is the Helmholtz free energy.

We can compute the *average magnetization* $M(\vec{x})$ as

$$M(\vec{x}) = \langle m(\vec{x}) \rangle \quad (117)$$

$$= \frac{1}{Z[h]} \int Dm m(\vec{x}) e^{-\beta \mathcal{H}_{\text{eff}}[m] + \beta \int d^d x h(\vec{x}) m(\vec{x})} \quad (118)$$

$$= \frac{1}{\beta Z[h]} \frac{\delta Z[h]}{\delta h(\vec{x})} \quad (119)$$

$$= \frac{1}{\beta} \frac{\delta \log Z[h]}{\delta h(\vec{x})} = -\frac{\delta F[h]}{\delta h(\vec{x})}. \quad (120)$$

Physical observables via functional derivatives

The *two-point correlation function* $G(\vec{x}, \vec{y})$ is defined by

$$G(\vec{x}, \vec{y}) = \langle (m(\vec{x}) - M(\vec{x}))(m(\vec{y}) - M(\vec{y})) \rangle \quad (121)$$

$$= \langle (m(\vec{x})m(\vec{y})) \rangle - M(\vec{x})M(\vec{y}) \quad (122)$$

$$\begin{aligned} &= \frac{1}{\beta^2 Z[h]} \frac{\delta^2 Z[h]}{\delta h(\vec{x}) \delta h(\vec{y})} - \left(\frac{1}{\beta Z[h]} \frac{\delta Z[h]}{\delta h(\vec{x})} \right) \left(\frac{1}{\beta Z[h]} \frac{\delta Z[h]}{\delta h(\vec{y})} \right) \\ &= \frac{1}{\beta} \frac{\delta}{\delta h(\vec{y})} \left(\frac{1}{\beta Z[h]} \frac{\delta Z[h]}{\delta h(\vec{x})} \right) \end{aligned} \quad (123)$$

$$= \frac{1}{\beta} \frac{\delta M(\vec{x})}{\delta h(\vec{y})} \equiv k_B T \chi_T(\vec{x}, \vec{y}) \quad (124)$$

where $\chi_T(\vec{x}, \vec{y})$ is the *isothermal susceptibility* and the equality represents a weak version of *fluctuation-dissipation theorem*.

For translational invariant systems, the *isothermal susceptibility* becomes

$$\chi_T = \lim_{|\vec{k}| \rightarrow 0} \hat{\chi}_T(\vec{k}) = \beta \lim_{|\vec{k}| \rightarrow 0} \hat{G}(\vec{k}) = \beta \int d\vec{r} G(\vec{r}). \quad (125)$$

Physical observables via functional derivatives

Legendre transform of the (Helmholtz) free energy is the *Gibbs free energy*

$$\tilde{F}[M] = F[h] + \int d\vec{y} h(\vec{y}) M(\vec{y}) \quad (126)$$

with $h(\vec{x}) = h[\vec{x}; M(\vec{x})] = h[M]$ specified by the conditions

$$0 = \frac{\delta F[h]}{\delta M(\vec{x})} = \frac{\delta}{\delta M(\vec{x})} \left(\tilde{F}[M] - \int d\vec{y} h(\vec{y}) M(\vec{y}) \right) \implies h(\vec{x}) = \frac{\delta \tilde{F}[M]}{\delta M(\vec{x})} \quad (127)$$

and conversely,

$$\frac{\delta \tilde{F}[M]}{\delta h(\vec{x})} = \frac{\delta F[h]}{\delta h(\vec{x})} + M(\vec{x}) = 0 \implies M(\vec{x}) = -\frac{\delta F[h]}{\delta h(\vec{x})}. \quad (128)$$

EXERCISE: Prove it applying the chain rule to the definition of $\tilde{F}[M]$.

REMARK: The difference between $\tilde{F}[M]$ and $\mathcal{H}_{\text{eff}}[m]$ is in the saddle-point operation.

Physical observables via functional derivatives

From the identity

$$\delta(\vec{x} - \vec{y}) = \frac{\delta h(\vec{x})}{\delta h(\vec{y})} \quad (129)$$

$$= \int d\vec{z} \frac{\delta h(\vec{x})}{\delta M(\vec{z})} \frac{\delta M(\vec{z})}{\delta h(\vec{y})} \quad (130)$$

$$= \beta \int d\vec{z} \frac{\delta h(\vec{x})}{\delta M(\vec{z})} G(\vec{z} - \vec{y}), \quad (131)$$

the inverse of the two-point correlation function can be defined as

$$\Gamma(\vec{x} - \vec{y}) \equiv G^{-1}(\vec{x} - \vec{y}) \quad (132)$$

$$= \beta \frac{\delta h(\vec{x})}{\delta M(\vec{y})} \quad (133)$$

$$= \beta \frac{\delta^2 \tilde{F}[M]}{\delta M(\vec{x}) \delta M(\vec{y})}. \quad (134)$$

Saddle-point and linear response

Applying saddle-point condition to the LG effective Hamiltonian,

$$0 = \frac{\delta}{\delta m(\vec{x})} \left\{ \int d\vec{y} \left[\frac{c_0}{2} (\nabla m)^2 + \frac{r_0 t}{2} m(\vec{y})^2 + \frac{u_0}{4} m(\vec{y})^4 \right] - \beta \int d\vec{y} h(\vec{y}) m(\vec{y}) \right\} \Big|_{m=M} \quad (135)$$

implies that the equilibrium state solves the partial differential equation,

$$-c_0 \nabla^2 M(\vec{x}) + r_0 t M(\vec{x}) + u_0 M(\vec{x})^3 - \beta h(\vec{x}) = 0, \quad (136)$$

with specified boundary conditions.

For zero or constant external field, in a translational invariant system, the equilibrium state is the homogeneous solution $M(\vec{x}) = M_0, \forall \vec{x}$.

Saddle-point and linear response

Linear response regime:

perturb the homogeneous state with a small localized field $\delta h(\vec{y})$, i.e.

$$\frac{\delta}{\delta h(\vec{y})} \left[-c_0 \nabla^2 M(\vec{x}) + r_0 t M(\vec{x}) + u_0 M(\vec{x})^3 - \beta h(\vec{x}) \right] \Big|_{h=0} = 0 \quad (137)$$

gives

$$\beta \delta(\vec{x} - \vec{y}) = \beta \frac{\delta h(\vec{x})}{\delta h(\vec{y})} \quad (138)$$

$$= \left(-c_0 \nabla^2 + r_0 t + 3u_0 M_0^2 \right) \frac{\delta M(\vec{x})}{\delta h(\vec{y})} \quad (139)$$

$$= \beta \left(-c_0 \nabla^2 + r_0 t + 3u_0 M_0^2 \right) G(\vec{x} - \vec{y}) \quad (140)$$

Saddle-point and linear response

Defining a *correlation length*

$$\xi = \begin{cases} \xi_{>} = \left(\frac{c_0}{r_0 t}\right)^{1/2}, & M_0 = 0, \quad \text{for } t > 0 \\ \xi_{<} = \left(-\frac{c_0}{2r_0 t}\right)^{1/2}, & M_0 = \pm\sqrt{-r_0 t/u_0}, \quad \text{for } t \leq 0 \end{cases} \quad (141)$$

we obtain ($c_0 = \bar{c}_0 k_B T$)

$$\left(-\nabla^2 + \xi^{-2}\right) G(\vec{x} - \vec{y}) = \frac{k_B T}{\bar{c}_0} \delta(\vec{x} - \vec{y}) \quad (142)$$

with

$$\xi \sim (T - T_0)^{-\nu}, \quad \nu = \nu_{MF} = \frac{1}{2}. \quad (143)$$

Saddle-point and linear response

In the Fourier space,

$$\left(-\nabla^2 + \xi^{-2}\right) \int \frac{d\vec{k}}{(2\pi)^d} \hat{G}(\vec{k}) e^{i\vec{k}\cdot(\vec{x}-\vec{y})} = \frac{k_B T}{\bar{c}_0} \int \frac{d\vec{k}}{(2\pi)^d} e^{i\vec{k}\cdot(\vec{x}-\vec{y})} \quad (144)$$

$$\int \frac{d\vec{k}}{(2\pi)^d} \hat{G}(\vec{k}) \left(k^2 + \xi^{-2}\right) e^{i\vec{k}\cdot(\vec{x}-\vec{y})} = \int \frac{d\vec{k}}{(2\pi)^d} \frac{k_B T}{\bar{c}_0} e^{i\vec{k}\cdot(\vec{x}-\vec{y})} \quad (145)$$

$$\implies \hat{G}(\vec{k}) = \frac{k_B T}{\bar{c}_0} \frac{1}{k^2 + \xi^{-2}}, \quad (146)$$

which is the *Ornstein-Zernicke form* of the correlation function.

Saddle-point and linear response

Inverting this expression,

$$G(\vec{r}) = \frac{k_B T}{\bar{c}_0} \int \frac{d\vec{k}}{(2\pi)^d} \frac{e^{i\vec{k}\cdot\vec{r}}}{k^2 + \xi^{-2}} \quad (147)$$

$$= \frac{k_B T}{\bar{c}_0 (2\pi)^d} \int_0^\Lambda dk k^{d-1} \int_\Omega d\Omega(\theta) \frac{e^{ikr \cos \theta}}{k^2 + \xi^{-2}} \quad (148)$$

$$= \frac{k_B T r^{2-d}}{\bar{c}_0 (2\pi)^d} \int_0^\Lambda dz z^{d-1} \int_\Omega d\Omega(\theta) \frac{e^{iz \cos \theta}}{z^2 + \left(\frac{r}{\xi}\right)^2} \quad (149)$$

$$\propto r^{2-d} Y\left(\frac{r}{\xi}\right) \quad (150)$$

where $Y(x)$ is a *scaling function*, with different asymptotic behavior for $r \ll \xi$ and $r \gg \xi$.

Saddle-point and linear response

For instance, in $d = 3$, we can explicitly compute

$$G(r) = \frac{k_B T r^{-1}}{\bar{c}_0 (2\pi)^3} \int_0^\infty dz z^2 \int_{-\pi}^\pi d\theta \sin \theta \int_0^{2\pi} d\phi \frac{e^{iz \cos \theta}}{z^2 + \left(\frac{r}{\xi}\right)^2} \quad (151)$$

$$= \frac{k_B T}{\bar{c}_0 8\pi^3 r} \int_0^\infty dz \frac{z^2}{z^2 + \left(\frac{r}{\xi}\right)^2} 2\pi \int_{-1}^1 d \cos \theta e^{iz \cos \theta} \quad (152)$$

$$= \frac{k_B T}{\bar{c}_0 2\pi^2 r} \int_0^\infty dz \frac{z \sin z}{z^2 + \left(\frac{r}{\xi}\right)^2} \quad (153)$$

$$= \frac{k_B T}{\bar{c}_0} \frac{1}{4\pi r} e^{-r/\xi}, \quad (154)$$

Saddle-point and linear response

More generally, writing the Laplacian operator in spherical coordinates,

$$\left[-\frac{1}{r^{d-1}} \frac{\partial}{\partial r} r^{d-1} \frac{\partial}{\partial r} + \xi^{-2} \right] G(r) = \frac{k_B T}{\bar{c}_0} \delta(r) \quad (155)$$

and performing rescaling $\rho = r/\xi$, we get

$$\left[-\frac{1}{\rho^{d-1}} \frac{\partial}{\partial \rho} \rho^{d-1} \frac{\partial}{\partial \rho} + 1 \right] G(\rho) = \frac{k_B T}{\bar{c}_0} \xi^{d-2} \delta(\rho), \quad (156)$$

with solution

$$G(r) = \begin{cases} \frac{k_B T}{\bar{c}_0} \xi^{2-d} e^{-r/\xi} & d = 1 \\ \frac{k_B T}{\bar{c}_0 (2\pi)^{d/2}} \xi^{1-d/2} r^{\frac{2-d}{2}} K_{\frac{d-2}{2}}(r/\xi) & d \geq 2, \end{cases} \quad (157)$$

where $K_\nu(x)$ is the modified spherical Bessel function of the third kind.

Saddle-point and linear response

$$K_\nu(x) \sim \begin{cases} \left(\frac{\pi}{2x}\right)^{\frac{1}{2}} e^{-x} & x \rightarrow \infty \\ \frac{\Gamma(\nu)}{2} \left(\frac{x}{2}\right)^{-\nu} & x \rightarrow 0, \nu > 0 \\ -\log x & \nu = 0, x \rightarrow 0. \end{cases} \quad (158)$$

The two-point correlation function takes the asymptotic behavior

- for $T \neq T_0$, $r \gg \xi$, $d \geq 2$,

$$G(r) \propto \frac{\pi^{(1-d)/2}}{2^{(1+d)/2}} \frac{k_B T}{\bar{c}_0} \xi^{\frac{3-d}{2}} \frac{1}{r^{\frac{d-1}{2}}} e^{-r/\xi} \propto \frac{1}{r^{\frac{d-1}{2}}} e^{-r/\xi} \quad (159)$$

- for $T = T_0$, $\xi \rightarrow \infty$, $d > 2$,

$$G(r) = \frac{\Gamma\left(\frac{d-1}{2}\right)}{4\pi^{d/2}} \frac{k_B T}{\bar{c}_0} \frac{1}{r^{d-2}} \propto r^{2-d} \quad (160)$$

which is of the form $r^{2-d-\eta}$ with $\eta = \eta_{MF} = 0$.